Glio Core

PET · MRI · GLIOMAANALYSIS

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Validation Report

Clustering benchmark on BraTS 2024 (MRI) — full dataset

271 cases · 5 methods · confidence intervals and paired tests

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1. Executive summary

This document reports the validation of GlioCore's clustering algorithms on the entire training_data_additional subset of BraTS 2024 (271 cases), in MRI mode. Five methods — a threshold baseline and four clustering methods — are compared with a protocol that avoids optimizing the metrics on the ground truth, with statistical analysis based on 95% confidence intervals and paired Wilcoxon tests.

Main outcome: a characterized trade-off, with no absolute winner

No method prevails across all metrics. The proposed hierarchical approach achieves the highest structural agreement (ARI 0.52) at the lowest computational cost among the non-trivial methods, being statistically equivalent to MRF-EM ($p = 0.082$) but about 17 times faster.

On enhancing-tumor segmentation, however, GMM and MRF-EM lead (Dice ET ≈ 0.73 versus 0.66 for the hierarchical method). The choice of method therefore depends on the goal: global structural agreement and efficiency point to the hierarchical method; maximum enhancing-tumor precision points to GMM.

The claims of this report are scoped to clustering validation in MRI mode. Metabolic PET segmentation, the conceptual core of GlioCore, is not quantitatively validated in this study.

2. Scope and limitations of the study

2.1 What the study demonstrates

• The behavior of the clustering algorithms on real multi-contrast MRI data, on the entire available dataset (271 cases).
• A statistically grounded comparison of five methods on the same cohort, with confidence intervals and paired tests.
• The accuracy/efficiency/subregion trade-off profile of each method.

2.2 What the study does not demonstrate

• The validity of metabolic PET segmentation, GlioCore's conceptually distinctive feature, which requires expert annotations on PET data not available in this study.
• Superiority over state-of-the-art supervised methods, not included in the comparison.
• The added value of the hybrid PET+MRI modality, whose validation requires complete multimodal cohorts.

3. Dataset and protocol

3.1 Dataset

The entire training_data_additional subset of BraTS 2024 (271 cases) was used, with the labeling conventions of the main dataset: each case provides four co-registered MRI contrasts (T1, T1ce, T2, FLAIR) and an expert manual segmentation. Regions: Whole Tumor = 1+2+3; Tumor Core = 1+3; Enhancing Tumor = 3.

3.2 Segmentation across all available channels

Distinctive feature: multivariate features across all channels

In MRI mode GlioCore uses all four contrasts (T1, T1ce, T2, FLAIR) simultaneously as the feature space; in PET+MRI the same logic combines up to six channels (SUVR, SUV, T1, T1ce, T2, FLAIR). Most clustering tools operate on a single intensity channel: segmentation across the full available multi-channel space, with automatic adaptation to the channels present, is a feature implemented and active in this benchmark.

3.3 Methods compared

Metodo	Type	Role in the comparison
Threshold	Quantile threshold on the primary feature	Baseline / minimum reference
FCM	Fuzzy C-Means	Soft clustering
GMM	Gaussian Mixture (BIC)	Parametric clustering
MRF-EM	GMM + Ising spatial prior	Clustering with spatial coherence
Hierarchical	Two-level hierarchical (proposed)	Original GlioCore method

3.4 Computation protocol

- Segmentation within the tumor mask (seg > 0): internal subdivision is evaluated, not detection. Whole Tumor Dice is ≈1 by construction and excluded from the metrics.
- Cluster-region mapping fixed a priori by intensity order (enhancing = highest-intensity cluster), without consulting the ground truth.
- Regions evaluated only in cases where they exist in the ground truth (233 of 271 cases for TC and ET).
- Metrics: Dice, Jaccard, HD95 on the subregions; ARI on the full partition; runtime. 95% confidence intervals via bootstrap (10,000 resamples).

4. Results

4.1 Comparative table with confidence intervals

Median values over 271 cases (ARI) and 233 cases (Dice TC/ET), with 95% confidence interval.

Metodo	ARI [95% CI]	Dice ET [95% CI]	Dice TC [95% CI]	Runtime
Threshold	0.13 [0.11–0.15]	0.55 [0.49–0.61]	0.36 [0.30–0.40]	1.6 s
FCM	0.24 [0.21–0.27]	0.38 [0.32–0.43]	0.30 [0.25–0.35]	4.3 s
GMM	0.40 [0.36–0.42]	0.74 [0.67–0.76]	0.41 [0.37–0.45]	8.0 s
MRF-EM	0.41 [0.37–0.46]	0.73 [0.67–0.77]	0.40 [0.34–0.44]	73 s
Hierarchical	0.52 [0.48–0.56]	0.66 [0.59–0.72]	0.28 [0.24–0.31]	4.4 s

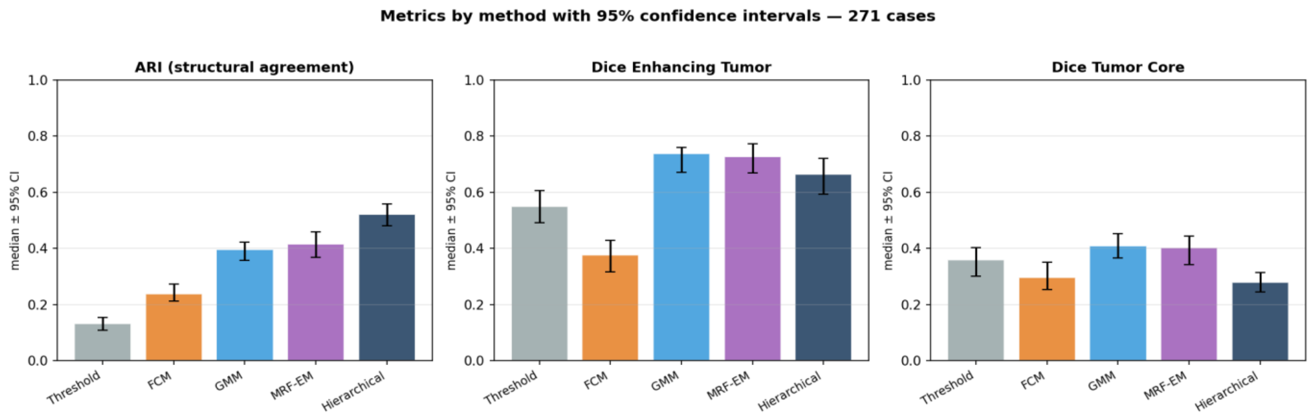

Figure 1 — Metrics by method with 95% confidence intervals (271 cases). Hierarchical leads on ARI; GMM and MRF-EM on enhancing; Tumor Core remains low for all.

4.2 ARI distribution

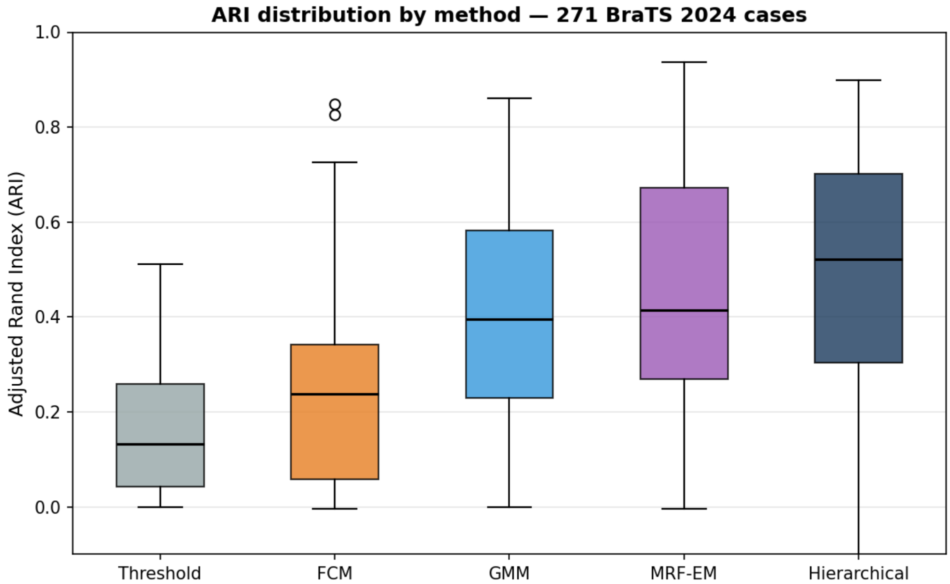

Figure 2 — ARI distribution by method. The median grows from the baseline to the hierarchical method; the Hierarchical and MRF-EM distributions overlap widely, consistent with the absence of a significant difference.

4.3 Paired statistical tests (ARI)

Paired Wilcoxon test (signed-rank, alternative hypothesis: Hierarchical superior) on the 271 cases, with effect size (Cliff's delta).

Comparison	p-value	Cases won	Cliff's δ	Outcome
Hierarchical vs Threshold	< 0.000001	246/271	0.84	Significant, large effect
Hierarchical vs FCM	< 0.000001	210/271	0.58	Significant, large effect
Hierarchical vs GMM	< 0.000001	169/271	0.27	Significant, small-medium effect
Hierarchical vs MRF-EM	0.082	135/271	0.02	Not significant

Reading the tests

The hierarchical method outperforms the baseline, FCM and GMM in a statistically significant way on ARI. Versus MRF-EM the difference is not significant (p = 0.082, negligible effect): the two methods are equivalent in structural agreement. Since the hierarchical method achieves this in 4.4 seconds against 73 for MRF-EM, its distinctive contribution is efficiency at equal structural accuracy.

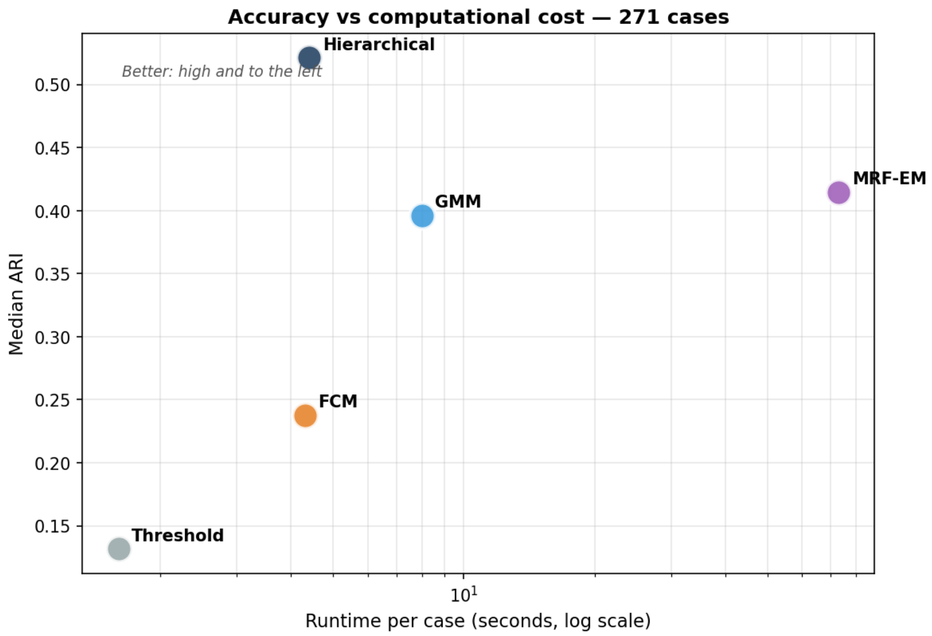

Figure 3 — Accuracy (ARI) against computational cost (logarithmic scale). Hierarchical and MRF-EM achieve equivalent ARI, but with runtimes separated by a factor of ~17.

5. Discussion

5.1 A trade-off, not a single winner

The full dataset reveals a nuanced picture. In terms of overall structural agreement (ARI), the hierarchical method is the best and ties MRF-EM at much lower cost. In terms of the single enhancing subregion (Dice ET), GMM and MRF-EM prevail. The Tumor Core remains low for all methods. There is therefore no dominant method: each presents a distinct strength profile, and the choice depends on the clinical or research objective.

5.2 The inductive bias of the hierarchical method

A note on interpreting the ARI

The hierarchical method is designed to produce a three-level partition that mirrors the hierarchical structure of the BraTS classes. Since the ARI rewards agreement with that structure, the method enjoys a favorable inductive bias relative to flat clustering methods. This does not invalidate the result, alignment with the structure of the problem is a desirable property, but it circumscribes its generality.

The fact that the advantage on ARI is clear while that on enhancement is absent is consistent with this reading: the method excels where the metric rewards the hierarchical structure for which it was designed.

5.3 The Tumor Core

Tumor Core Dice is low for all methods (0.28–0.41). On structural MRI, necrosis and edema present partly overlapping signals, limiting separation for any unsupervised method. The hierarchical method, which emphasizes the enhancing/non-enhancing distinction, shows the lowest value here: it favors the separation it captures best (enhancement) over the necrosis/edema one. The Tumor Core remains the point where a robust contribution will have to prove itself, primarily through the PET component and active learning.

6. Methodological verification

The results were subjected to independent checks on all 1,355 records (271 cases × 5 methods).

Check	Outcome
All 271 cases processed without errors, across all methods	Passed
Dice and Jaccard within [0,1] on all records	Passed
Jaccard ≤ Dice relation respected everywhere	Passed
ARI within the valid range	Passed
WT sanity = 1.0 (consistent with segmentation within seg>0)	Passed
Independent verification of the threshold baseline code	Passed

Transparency on a methodological correction

An initial version of the protocol selected the clusters to merge by maximizing Dice against the ground truth, constituting a metric optimization that inflated its values. The protocol was corrected by adopting an a-priori intensity mapping, independent of the ground truth. The values reported here derive from the corrected protocol, applied uniformly to all methods.

7. Limitations

- Quantitative validation covers only the clustering algorithms in MRI mode; metabolic PET segmentation is not quantitatively validated due to the absence of expert annotations on PET data.
- The comparison includes five unsupervised methods but no state-of-the-art supervised method.
- The hierarchical method's advantage on ARI is partly attributable to its favorable inductive bias relative to the BraTS structure.
- Segmentation operates within the provided tumor mask; tumor localization is not evaluated.
- On structural MRI, necrosis/edema separation limits the Tumor Core for all unsupervised methods.

8. Future work

1. Quantitative PET validation: expert annotations on PET patients to validate metabolic segmentation, extending validation to the conceptual core of the project.

2. PET+MRI validation: the hybrid modality is implemented; its validation requires cohorts with co-registered, annotated PET and multi-contrast MRI. The key experiment will be to verify whether the metabolic channels improve accuracy over MRI alone.

3. Comparison con metodi supervisionati allo stato dell'arte, per collocare i risultati nel contesto del dominio.

4. Ablation analysis on channels and hierarchical levels, to quantify the contribution of each component.

5. Active learning on borderline necrosis/edema cases, where unsupervised clustering shows its limits.

Summary

On the full BraTS 2024 dataset, GlioCore provides a statistically grounded clustering comparison in MRI mode. The proposed hierarchical method offers the best overall structural agreement at minimal computational cost, equivalent to the best competitor but much faster, while maximum enhancing-tumor precision remains the domain of GMM and MRF-EM. The claims are scoped to clustering validation in MRI; consolidation toward an overall demonstration of the project requires quantitative validation of the PET component and of the cross-modal value.

9. Bibliography

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